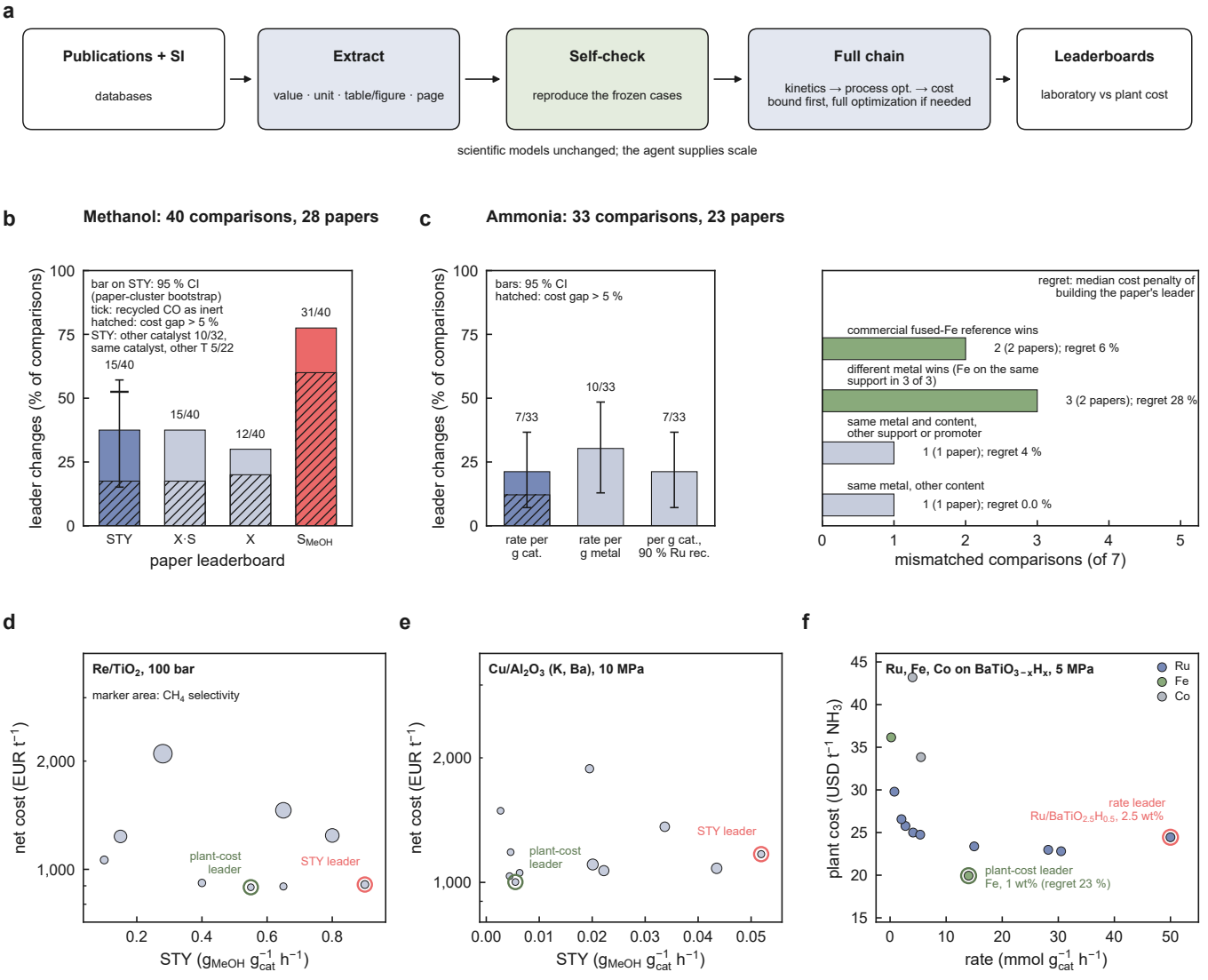

a
Publications + SI
databases
Extract
value · unit · table/figure · page
Self-check
reproduce the frozen cases
Full chain
kinetics → process opt. → cost
bound first, full optimization if needed
Leaderboards
laboratory vs plant cost
scientific models unchanged; the agent supplies scale
b
Methanol: 40 comparisons, 28 papers
bar on STY: 95 % CI
(paper-cluster bootstrap)
tick: recycled CO as inert
hatched: cost gap > 5 %
STY: other catalyst 10/32,
same catalyst, other T 5/22
15/40
15/40
12/40
31/40
leader changes (% of comparisons)
STY
X·S
X
S_MeOH
paper leaderboard
c
Ammonia: 33 comparisons, 23 papers
bars: 95 % CI
hatched: cost gap > 5 %
7/33
10/33
7/33
rate per g cat.
rate per g metal
per g cat., 90 % Ru rec.
regret: median cost penalty of building the paper's leader
commercial fused-Fe reference wins
2 (2 papers); regret 6 %
different metal wins (Fe on the same support in 3 of 3)
3 (2 papers); regret 28 %
same metal and content, other support or promoter
1 (1 paper); regret 4 %
same metal, other content
1 (1 paper); regret 0.0 %
mismatched comparisons (of 7)
d
Re/TiO2, 100 bar
marker area: CH4 selectivity
net cost (EUR t−1)
plant-cost leader
STY leader
STY (gMeOH gcat−1 h−1)
e
Cu/Al2O3 (K, Ba), 10 MPa
net cost (EUR t−1)
plant-cost leader
STY leader
STY (gMeOH gcat−1 h−1)
f
Ru, Fe, Co on BaTiO3−xHx, 5 MPa
Ru
Fe
Co
plant cost (USD t−1 NH3)
rate leader
Ru/BaTiO2.5H0.5, 2.5 wt%
plant-cost leader
Fe, 1 wt% (regret 23 %)
rate (mmol gcat−1 h−1)